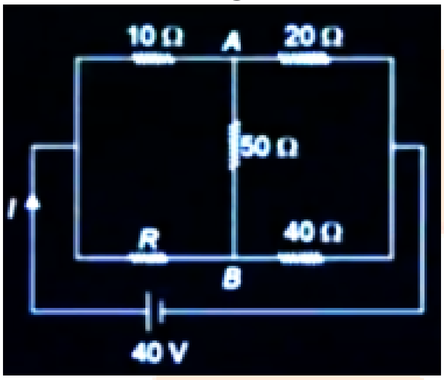
JEE-Main-28-01-2025 (Memory Based)
[EVENING SHIFT]

Physics

Question: In the given circuit, find I if the potentials at A and B are equal



Options:
(a) 1 A
(b) 2 A
(c) 3 A
(d) 4 A
Answer: (b)

Question: Bohr's model is applicable for a single electron atom of atomic number Z. Dependency of frequency of rotation of electron in n^{th} principal quantum number is proportional to
Options:
(a) Z/n^2
(b) Z^2/n^3
(c) n^3/Z
(d) Z/n
Answer: (b)

Question: In an electromagnetic wave, the magnetic field is given as
$\vec{B} = \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right) 30 \sin(\omega t - kz)$ **the corresponding electric field is**
Options:
(a) $\left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right) 9 \times 10^9 \sin(\omega t - kz)$
(b) $\left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right) 9 \times 10^9 \sin(\omega t - kz)$

(c) $\left(\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2}\hat{j}\right)9\times10^{9}\cos(\omega t-kz)$

(d) $\left(\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}\hat{j}\right)9\times10^{9}\cos(\omega t-kz)$

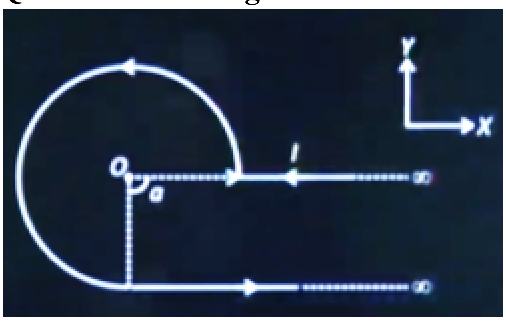
Answer: (b)

Question: For concave mirror, distance between object and image = 20cm and m = - 3 find focal length
Options:
(a) -7.5 cm
(b) -15 cm
(c) -20 cm
(d) -10 cm
Answer: (a)

Question: Wave theory of light cannot explain
Options:
(a) Compton effect
(b) Reflection of light
(c) Refraction of light
(d) Diffraction of light
Answer: (a)

Question: The mass and radius of a planet P is 8 and 3 times that of earth respectively. If escape velocity from surface of earth is V_e , then escape velocity from surface of planet P is
Options:
(a) $\sqrt{\frac{8}{3}}V_e$
(b) $\sqrt{24}\ V_e$
(c) $\sqrt{\frac{3}{8}}V_e$
(d) $8/3\ V_e$
Answer: (a)

Question: The magnetic field $\vec{B}$ at the centre O of the given arrangement is



Options:

(a) $\frac{+\mu_0 l}{8\pi a}(3\pi + 2)\widehat{k}$

(b) $\frac{-\mu_0 l}{8\pi a}(3\pi + 2)\widehat{k}$

(c) $\frac{+\mu_0 l}{8\pi a}(3\pi - 2)\widehat{k}$

(d) $\frac{-\mu_0 l}{8\pi a}(3\pi - 2)\widehat{k}$

Answer: (a)

Question: A cube of side 10 cm having bulk modulus of 1.4 x 1011 Pa is placed in the atmosphere. Now it is subjected to extra pressure of 7x 10^6 Pa then magnitude of change in volume of cube is

Options:

(a) 0.03 mL

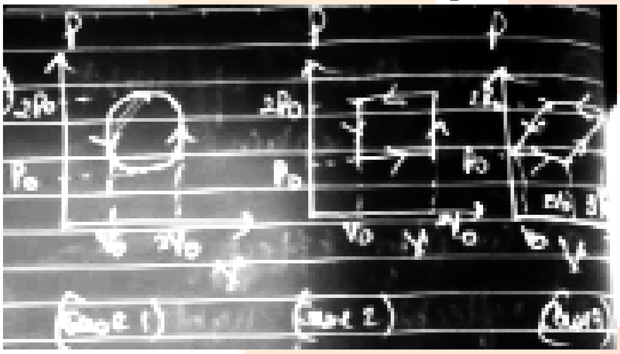
(b) 0.3 mL

(c) 0.05 mL

(d) 0.2 mL

Answer: (c)

Question: What is the relationship between change in internal energy of each case ?



Options:

(a) $\Delta U_1 > \Delta U_2 > \Delta U_3$

(b) $\Delta U_1 = \Delta U_2 = \Delta U_3$

(c) $\Delta U_1 < \Delta U_2 < \Delta U_3$

(d) $\Delta U_1 = \Delta U_2 \neq \Delta U_3$

Answer: (b)

Question: A parallel plate capacitor of capacitance 6 μF is charged by a battery of voltage 10 V. Area of plate 10 cm^2 . Find energy density

Options:

(a) $\frac{18}{\varepsilon_0} \times 10^{-7}$

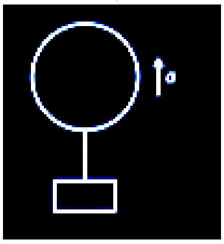
(b) $\frac{9}{\varepsilon_0} \times 10^{-7}$

(c) $\frac{25}{\varepsilon_0} \times 10^{-8}$

(d) $\frac{18}{\varepsilon_0} \times 10^{-8}$

Answer: (d)

Question: A balloon system having mass m is moving up with acceleration a, find the mass to be removed from it to have acceleration 3a. (Neglect the volume of mass attached)



Options:

(a) $\frac{2ma}{3a+g}$

(b) $\frac{2ma}{2a+g}$

(c) $\frac{ma}{3a+g}$

(d) $\frac{ma}{g-3a}$

Answer: (a)

Question: An equilateral triangle frame of side l is carrying current i, find magnetic field at its centroid

Options:

(a) $\frac{3\mu_0 l}{4\pi l}$

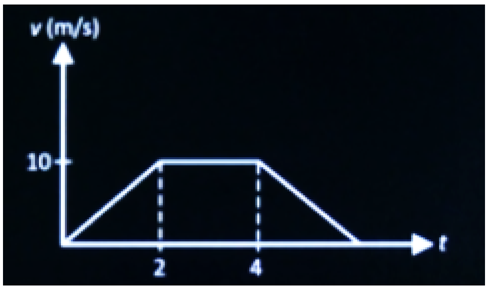
(b) $\frac{3\mu_0 l}{\pi l}$

(c) $\frac{9\mu_0 l}{2\pi l}$

(d) $\frac{\mu_0 l}{\pi l}$

Answer: (c)

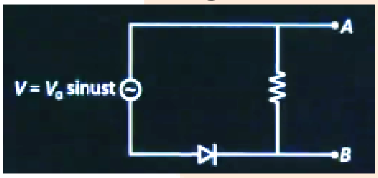
Question: The velocity vs time graph of a particle moving along X-axis is plotted as shown. The distance travelled (in metre) by the particle in the interval t = 0 s to t = 4 s is

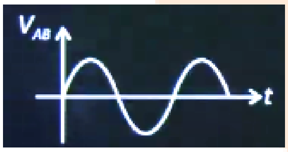


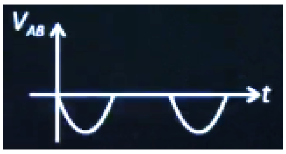
Options:
(a) 10
(b) 20
(c) 30
(d) 40
Answer: (c)

Question: The translational Kinetic energy of molecules of 50g of CO_2 gas at 17°C is
Options:
(a) 2500J
(b) 4110 J
(c) 5250 J
(d) 6300 J
Answer: (b)

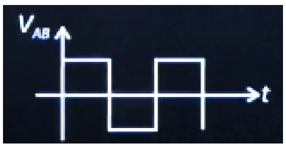
Question: The correct variation of voltage across AB is given by (consider that the threshold voltage of the diode is very small)



Options:
(a) 

(b) 

(c) 

(d) 

Answer: (b)

Question: An electric dipole of moment 6×10^{-6} cm is placed parallelly in electric field of strength 10^6 N/C. Work done required to rotate the dipole by 180° is X joules, then X is
Options:
(a) 5
(b) 20
(c) 18
(d) 12
Answer: (d)

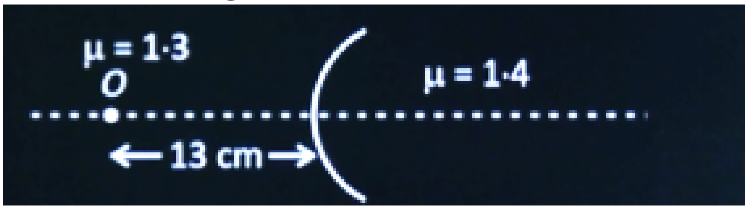
Question: Distance between real object and its three times magnified image formed by concave mirror is 20 cm then radius of curvature of the mirror is X cm, then X is
Options:
(a) 15
(b) 10
(c) 5
(d) 25
Answer: (a)

Question: Select the correct match for dimensions

Column-I	Column-II
(A) Angular Momentum	(I) $[MLT^{-2}]$
(B) Force	(II) $[ML^2T^{-1}]$
(C) Energy	(III) $[ML^{-1}T^{-2}]$
(D) Pressure	(IV) $[ML^2T^{-2}]$

Options:
(a) A-(II), B(III), C-(I), D-(IV)
(b) A-(I), B(II), C-(III), D-(IV)
(c) A-(II), B(I), C-(IV), D-(III)
(d) A-(II), B(I), C-(III), D-(IV)
Answer: (c)

Question: In the figure shown the object kept at a distance 13 cm from the interface forms a real image which is double in size. The radius of curvature of the interface is



Options:
(a) 3/2 cm
(b) 2/3 cm

(c) 3/4 cm
(d) 4/3 cm
Answer: (b)

Question: Due to the bar magnet shown, if the % uncertainty in d is 1% , find uncertainty in the magnetic field at P. [d : 10 units, l = 10 units]
Options:
(a) 2%
(b) 3%
(c) 1.5 %
(d) 0.5 %
Answer: (c)

JEE-Main-28-01-2025 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: Consider the following oxides,
V_2O_3, V_2O_4 and V_2O_5
Change in oxidation state of vanadium when amphoteric oxide reacts with acids to form VO_4^+ is
Options:
(a) 1
(b) 2
(c) 3
(d) 4
Answer: (b)

Question: Bohr's model is applicable for single electron atom of atomic number Z. Dependency of frequency of rotation of electron in n^{th} principal quantum number is proportional to
Options:
(a) Z/n^2
(b) Z^2/n^3
(c) n^3/Z
(d) Z/n
Answer: (b)

Question: Which has maximum oxidizing power among the following
Options:
(a) VO_2^+
(b) $Cr_2O_7^{2-}$
(c) MnO_4^-
(d) TiO_2
Answer: (c)

Options: Calculate the spin magnetic moment of Mn_2O_3
(a) a = $\sqrt{24}$
(b) b = $\sqrt{36}$
(c) c = $\sqrt{34}$
(d) d = $\sqrt{20}$
Answer: (a)

Question: Which of the following compound(s) is/are yellow in colour?
(a) CdS, (b) PbS, (c) Cus, (d) ZnS (Cold), (e) $PbCrO_4$
Choose the correct answer from the options given below:
Options:
(a) (a), (c) and (e) only
(b) (a) and (e) only
(c) (b) and (d) only
(d) (a), (b) and (e) only

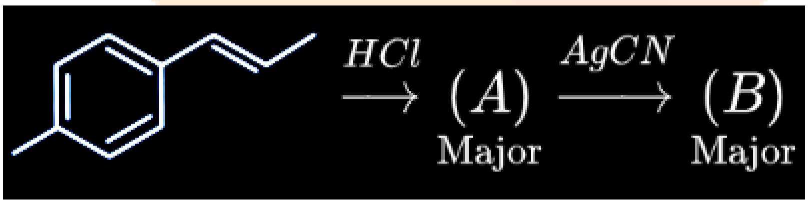
Answer: (b)

Question: $CH_3 - C{\equiv}CH \xrightarrow[H_2]{Pd/C} (A) \xrightarrow[(ii)Zn, H_2O]{(i)O_3} (B) + (C)$

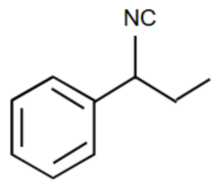
Options:
(a) B = CH3CHO
C = HCHO
(b) B = CH3CHO
C = HCOOH
(c) B=CH3COCH3
C= HCHO
(d) B ⇒ HCHO
C ⇒ CH_3COOH
Answer: (a)

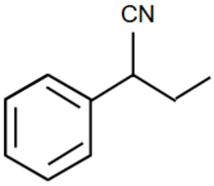
Question: The correct order of energy of the following subshell is
1s 2s 3p 3d
Options:
(a) 1s < 2s < 3d < 3p
(b) 2s < 1s < 3p < 3d
(c) 1s < 3p < 2s < 3d
(d) 1s < 2s < 3p < 3d
Answer: (d)

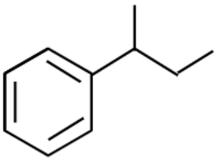
Question: Which has maximum oxidizing power among the following.
Options:
(a) VO_2^-
(b) $Cr_2O_7^{2-}$
(c) MnO_4^-
(d) TiO_2
Answer: (c)

Question: 

Options:

(a) 

(b) 

(c) 

(d) None of these

Answer: (a)

Question: Which of the group - 15 element forms dπ - dπ Bond and strongest basic hydride?

Options:

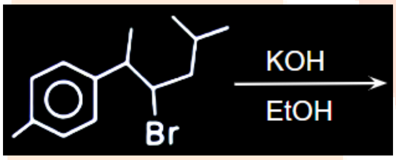
(a) Z = 7

(b) Z = 15

(c) Z = 33

(d) Z = 51

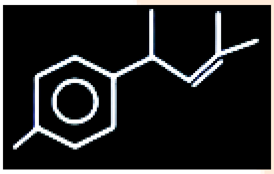
Answer: (b)

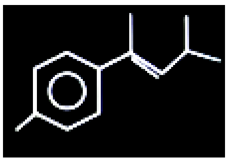
Question: 

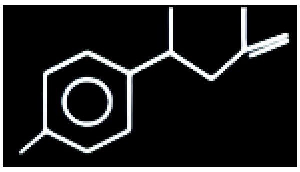
identify major

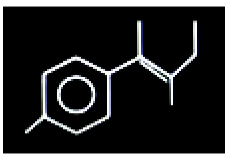
Product

Options:

(a) 

(b) 

(c) 

(d) 

Answer: (a)

Question: Which of the following complex is paramagnetic
Options:
(a) $[NiCl_4]^{2-}$
(b) $[Ni(CO)_4]$
(c) $[Ni(CN)_4]^{2-}$
(d) $[Fe(CO)_5]$
Answer: (a)

Question: 30 gm HNO3 is added to a solution to prepare 75% w/w solution having density 1.25 g/mL. Volume of solution is
Options:
(a) 32 mL
(b) 48 mL
(c) 36 mL
(d) 28 mL
Answer: (a)

Question: S-I 

are ring chain isomers

S-II 

are functional isomers
Options:
(a) Both S-I and S-II are correct Statements
(b) S-I is correct and S-II is not correct
(c) S-I wrong statement and S-II is correct statement
(d) Both Statements are correct
Answer: (a)

Question: For an elementary reaction
$A + B \rightarrow C + D$
When volume becomes ⅓ rd, rate of reaction becomes
Options:
(a) 8 times
(b) 9 times
(c) 6 times
(d) 2 times
Answer: (b)

Question: Match the following List-I with List-II

	List - I		**List-II**
A	**$[COF_6]^{3-}$**	**i**	**sp^3d^2**
B	**$[CO(NH_3)_6]^{3+}$**	**ii**	**d^2sp^3**

C	**$[NiCl_4]^{2-}$**	**iii**	**sp^3**
D	**$[Ni(CN)_4]^{2-}$**	**iv**	**dsp^2**

Choose the correct answer from the options given below:
Options:
(a) A-i, B-II, C-iii, D-iv
(b) A-ii, B-i, C-iv, D-iii
(c) A-i, B-ii, C-iv, D-iii
(d) A-ii, B-i, C-iii, D-iv
Answer: (a)

Question: The correct name of I & II in the following process is:
$Solid \rightarrow^{I} Vapours \rightarrow^{II} Solid$

Options:
(a) I → Sublimation
II → Vaporisation
(b) I → Sublimation
II → Decomposition
(c) I → Sublimation
II → Deposition
(d) I → Deposition
II → Sublimation
Answer: (c)

Question: Consider the following statements:
Statement I: In law of octaves, elements were arranged in increasing order of their atomic numbers.
Statement II: Lothar Meyer, plotted the physical properties against atomic weight
Choose the correct answer from the options given below:
Options:
(a) Both statement I and statement II are correct
(b) Both statement I and statement II are incorrect
(c) Statement I is correct but statement II is incorrect
(d) Statement I is incorrect but statement II is correct
Answer: (d)

Question: Consider the following sequence of reaction
$C_6H_{12} \rightarrow^{Se/\Delta} A \rightarrow_{AlC_3}^{CH_3-Cl} B \rightarrow_{H_3O^+}^{CrO_2Cl_2} C$
Choose the correct option about major product
Options:
(a) 'C' gives Fehling's solution test
(b) 'C' can be prepared by reacting PhMgBr with CO_2
(c) 'C' can give Tollen's test
(d) 'C' can give effervescence with $NaHCO_3$
Answer: (c)

Question: Which of the following biomolecules doesn't contain C_1 — C_4 glycosidic linkage
Options:
(a) Amylopectin
(b) Maltose
(c) Lactose
(d) Sucrose
Answer: (d)

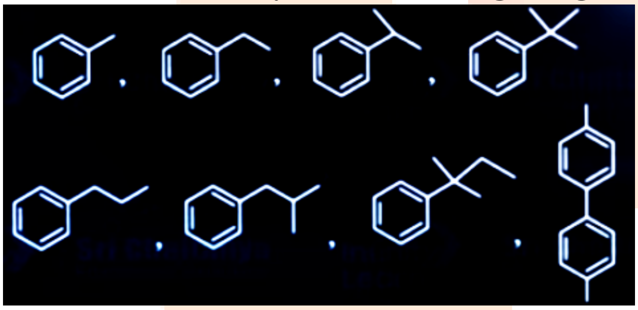
Question: No. of Paramagnetic species among the following is O_2, O_2^+, O_2^-, NO_2, NO, CO

Answer: (5)

Question: How many of the following molecules are polar?
CH_4, CCl_4 CH_2Cl_2, H_2O, NH_3, H_2O_2, O_2F_2

Answer: (5)

Question: How many of the following will give Benzoic acid on Oxidation with $KMnO_4$?



Answer: (6)

JEE-Main-28-01-2025 (Memory Based)
[EVENING SHIFT]
Maths

Question: Evaluate $\sum_{r=1}^{13} \frac{1}{sin\left[\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right]sin\left[\frac{\pi}{4}+\frac{r\pi}{6}\right]}$

Options:

(a) $2\sqrt{3} + 2$

(b) $2\sqrt{3} - 2$

(c) $3\sqrt{2} + 2$

(d) $3\sqrt{2} - 4$

Answer: (b)

$\sum_{r=1}^{13} \frac{1}{\sin\left[\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right]\sin\left[\frac{\pi}{4}+\frac{r\pi}{6}\right]}$

$\sum_{r=1}^{13} \frac{\sin\left[\frac{\pi}{4}+r\frac{\pi}{6}\right]-\left[\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right]}{\left(\sin\frac{\pi}{6}\right)\times\sin\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)\sin\left(\frac{\pi}{4}+\frac{\pi}{6}\right)}$

$\sum_{r=1}^{13} 2\left(\cot\left(\frac{\pi}{4}+(r-1)\frac{\pi}{6}\right)-\cot\left(\frac{\pi}{4}+\frac{r\pi}{6}\right)\right)$

$2\left[\cot\frac{\pi}{4}-\cot\left(\frac{\pi}{4}+\frac{13\pi}{6}\right)\right]$

$2\left[1-\cot\left(\frac{29\pi}{12}\right)\right]$

$2\left[1-\cot\left(2\pi+\frac{5\pi}{12}\right)\right]$

$2\left(1-\cot\frac{5\pi}{12}\right)$

$2\left(1-2+\sqrt{3}\right) \Rightarrow 2\left(\sqrt{3}-2\right)$

Question: $\int \frac{dx}{x^{\frac{1}{4}}\left(x^{\frac{1}{4}}+1\right)}$ **,f(0) = -6, f(2) = ?**

Solution :

$I = \int \frac{dx}{x^{1/4}\left(x^{1/4}+1\right)}, f(0) = -6, f(2)$

$I = \int \frac{4t^3dt}{t(t+1)}$

$= 4\int \frac{t^2}{t+1}dt$

$= 4\int \frac{t^2-1+1}{t+1}$

$= 4\int (t-1) + 4\int \frac{1}{t+1}$

$\Rightarrow 4\left(\frac{t^2}{2}-t\right)+4\ln(t+1)+c$

$f(x) = 2\left(x^{\frac{1}{2}}\right)-4x^{\frac{1}{4}}+4\ln\left(1+x^{\frac{1}{4}}\right)+c$

$C = -6$

$f(2) = 2\left(\sqrt{2}\right)-42^{\frac{1}{4}}+4\ln\left(1+2^{\frac{1}{2}}\right)$

Question: Bags B_1, B_2, B_3 contains 4 Blue, 6 white balls, 5 white 5 blue balls and 6 Blue 4 white balls respectively. A bag is randomly selected and a ball is drawn. If the drawn ball is white then find the probability that B_2 bag was selected.

Solution :

$B_1,\ B_2,\ B_3$

$B_1\ =\ +4\ \ Bl\ \ \&\ 6\ w$

$B_2 = -5\ \ wl\ \ \&\ \ 5\ \ Bl$

$B_3 = 5\ \ Bl\ \ \ \&\ \ \ 4\ \ Wl$

$P(B_2/A) \Rightarrow \frac{1/3\times 5/10}{1/3\times 5/10+1/3\times 6/10+1/3\times 4/10}$

$\Rightarrow \frac{\frac{5}{3\times 10}}{\frac{15}{3\times 10}} \Rightarrow \frac{1}{3}$

Question: Find domain of $\sec^{-1}(2[x] + 1)$, where [.] denotes GIF.
Solution :

Domain of $\text{Sec}^{-1}(2[x]+1)$

$2[x]+1 \geq 1$ on $2[x]\ +1 \leq -1$

$2[x]\ \geq\ 0 \qquad [x] \leq \frac{-2}{2}$

$x \geq\ 0 \qquad [x]\ \leq\ -1$

$[x]\ \leq\ -1$

$\therefore x\,t(-\infty,\ -1] \cup [0, \infty]$

Question: If $x^2 - (3 - 2i)x - (2i - 2) = 0$ has roots $\alpha + i\beta$ and $\gamma + i\delta$ find the value of $\alpha\gamma + \beta\delta$.
Solution :

$x^2\ -\ (3-2i)\,x\ -\ (2i\ -2)\ = 0$

$x = \frac{3-2i\pm\sqrt{9-4-12i+4(2i-2)}}{2}$

$x = \frac{(3-2i)\ \pm\sqrt{-4i-3}}{2}$

$= \left[\left(\frac{1}{\sqrt{2}}\right)^{10} - 2^{10}\right]\ =\ \sqrt{-4i-3}\ =\ \alpha - i\beta$

$-4i\ -\ 3\ = \alpha^2\ - \beta^2\ + i2\alpha\beta$

$x^2\ - y^2\ = -3\ \ \ 2\times\ y = 4$

$x^2\ - y^2\ = -3\ \ \ \alpha y\ = -2$

$x^2\ +\ y^2\ = \sqrt{\alpha^2\ - y^2\ + 4x^2y^2}\ = 5$

$x^2\ + y^2\ = 5$

$x^2\ - y^2 = -3$

$2x^2\ = 2 \rightarrow x^2\ = 1\ \Rightarrow\ \alpha = 1, \beta = -2$

$\alpha = -1, \beta = +2$

$n = \frac{(3-2i)\pm(1-2i)}{2}\ - 2i + 2$

$\alpha\ = 1,\ \ \beta = 0,\ \ \gamma = 2,\ \ \delta - 2$

$\alpha\gamma\ + \beta\delta\ = 2$

Question: $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & -2 \\ 0 & 1 \end{bmatrix} P = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ **$B = PAP^T$; $X = PB^{10}P^T$ Find X = ?**

Solution :

$A=\begin{bmatrix}\frac{1}{\sqrt{2}} & -2\\ 0 & 1\end{bmatrix} P=\begin{bmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{bmatrix}$

$PP^T=I \quad B=PAP^T \quad \mathrm{A}^2=\begin{bmatrix}\frac{1}{\sqrt{2}} & -2\\ 0 & 1\end{bmatrix}\begin{bmatrix}\frac{1}{\sqrt{2}} & -2\\ 0 & 1\end{bmatrix}$

$B^2=PAP^T(PAP^T) \quad A^2=\left[\left(\frac{1}{2}\right)^2 I^2\right]$

$=PA(P^TP)AP^T$

$B^2=PA^2P^T \qquad X=P^T(PA^{10}P^T)P$

$B^{10}=PA^{10}P^T \qquad =\mathrm{IA}^{10}I=A^{10}$

$x=P^TB^{10}P \quad \therefore x=\mathrm{A}^{10}=\begin{bmatrix}\left(\frac{1}{\sqrt{2}}\right)^{10} & -2^{10}\\ 0 & 1^{10}\end{bmatrix}$

Question: 212, 213,..........,999 find no. of numbers in the sequence above whose sum of digits is 15.

Solution :

$212, 213, \ldots\ldots\ldots\ldots, 999$

$\because Sum=15$

$x+y+2=15$

$x\in\{2,3,\ldots\ldots.9\}$

$y\in\{0,1,2,\ldots\ldots.9\}$

Coefficient $x^{15}:(x^2+x^3+\ldots..+n^9)(x^{11}+x^1+\ldots..x^9)^2$

$(1+x+x^2+\ldots\ldots+x^2)\left(\frac{1-x^{10}}{1-x}\right)^2$

$\left(\frac{1-x^8}{1-x}\right)\left(\frac{1-x^{10}}{1-x}\right)^2$

$(1-x^8)(1-x^{10})(1-x)^{-3}$

$(1-x^8)(1-2x^{10}+x^{20})(1-x)^{-3}$

$(1-x^8-2x^{10})(1-x)^{-3}$

Coefficient $x^{12}={}^{3+13-1}C_{13}-{}^{3+5-1}C_5-2^{3+3-1}C_3$

$=64$

Question: Words are formed using all letters of Word GARDEN. Find the probability in the word vowels are not together.

Solution :

GARDEN=6!

$\times G \times R \times D \times N \times$

$= {}^5C_2 \times 2! \times 4!$

$P(\varepsilon) = \frac{{}^5C_2 \times 2 \times 4!}{6!}$

$= \frac{20 \times 4!}{6 \cdot 5 \cdot 4!}$

$= \frac{2}{3}$

Question: Area bounded between the curves C_1 : $x(1+y^2)-1=0$ and C_2 : $y^2-2x=0$ is (in sq. unit)

Options:

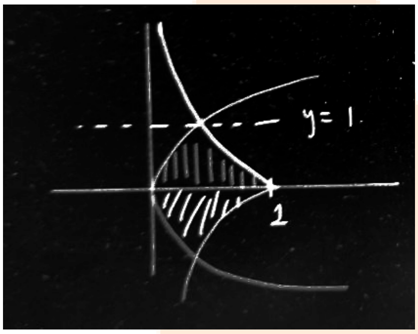
(a) $\frac{\pi}{2} - \frac{1}{3}$

(b) $\frac{\pi}{4} - \frac{1}{6}$

(c) $2\left(\frac{\pi}{2} - \frac{1}{6}\right)$

(d) $\frac{\pi}{6} + \frac{1}{2}$

Answer: (a)



$$C_x = x\left(1 + r^2\right) - 1 \text{ and}$$

$$C_y = y^2 - 2x = 0$$

$$\int_{-1}^{1} \left(\frac{1}{1+y^2} - \frac{y^2}{2} \right) dy$$

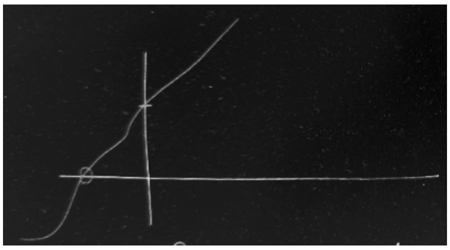
$$\left(\tan^{-1} y - \frac{y^3}{6} \right)_{-1}^{1}$$

$$\left(\frac{\pi}{4} - \frac{1}{6}\right) - \left(-\frac{\pi}{4} + \frac{1}{6}\right)$$

$$\frac{\pi}{2} - \frac{1}{3}$$

Question: If f(x) is polynomial satisfying $f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$ and f(2) = 129; then find real values of "k" satisfying f(k) = -2k.

Solution :



$$f(x) \cdot f\left(\tfrac{1}{x}\right) = f(x) + f\left(\tfrac{1}{x}\right)$$

$$f(x) = \pm x^4 + 1$$

$$f(x) = \pm 2^4 + 1 = 129$$

$$x = 7$$

$$f(x) = x^7 + 1$$

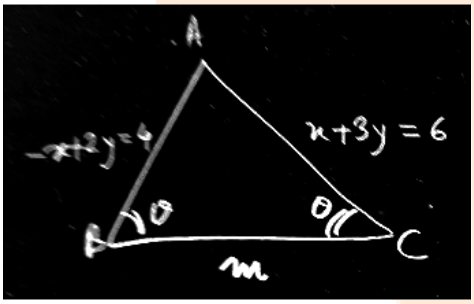
$$f(x) = k^7 + 1 = -2k$$

$$k^7 + 2k + 1 = 0$$

$$f'(k) = 7k^6 + 2$$

Question: An isosceles triangle formed by 3 lines. Equation of equal sides of triangle are -x + 2y = 4 & x + 3y = 6. Find sum of possible values of slope of 3rd side.

Solution :



$$\tan\theta = \left|\frac{m-\frac{1}{2}}{1+\frac{m}{2}}\right| = \left|\frac{\frac{-1}{2}-m}{1+\frac{m}{3}}\right|$$

$$\frac{2m-1}{m+2} = \pm\left(\frac{-1-3m}{3-m}\right)$$

$$\frac{2m-1}{m+2} = \frac{-(1+3m)}{3-m}$$

$$\frac{2m-1}{m+2} = \frac{3m+1}{3-m}$$